Cloud database technologies, database migration, document stores

Vassányi István
Pannon Egyetem, Veszprém

PaaS Data storage in the cloud—pros and cons

- ✓ high **availability** with a geographically distributed, fault tolerant infrastructure
- ✓ automated on-demand **scalability** with flexible, with (in part) load-dependent billing schemes
- ✓ cloud **infrastructure**, services like data analysis, image processing, NLP, AI, etc. which can be integrated in a solution
- ✗ services are harder to configure than the private servers, though some functions may be replaced by other means
- ✗ there are **fixed costs** to be paid for unused resources e.g. data storage. All such costs should be considered in a business plan.
- ✗ Data **security** may be an issue for highly sensitive data (like personal healthcare records
- ✗ **Critical services**, like production system control should not depend on Internet access -> **private cloud**

Database migration

- Between versions of the same vendor
  - Data types, obsolete features
  - -> automated solution (success depending on version difference)
- Between vendors, same architecture (e.g. MSSQL -> Postgres)
  - Data types, Syntax, Authentication, Stored logic
  - -> CSV (one table), wizards (trivial cases), dumps edited manually, third party tools, custom application
- Between architectures (e.g. relational -> document store)
  - All as above plus data remodelling
  - -> individual desing considerations based on application, custom application

DEMO: web app on GCP

Document stores

- noSQL = Not Only SQL
- Collections of semi-structured documents (json, xml, …)
- No schema, i.e. „schema-on-read”
- Relationships among entities
  - Embedding
  - Reference paths
- Strucure is application-specific

GCP Firestore

- Json based doc store
- Documents must have a key
- Security rules may enforce referential integrity
- Collections cannot be nested, but a document may hold a sub-collection (max depth 100) -> one-to-many relationships
- Limited basic data types, plus array, map, reference (path)
- Claims ACID type transactions, applicable for OLTP type backends

DEMO: NW as a doc store